WELDING AND FABRICATION

These notes introduce the science, equipment, procedures, quality control, safety, design considerations and practical workshop methods used in welding and metal fabrication. The material is written for classroom teaching, independent study and workshop preparation, with emphasis on engineering understanding rather than memorisation.

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1. Introduction to Welding and Fabrication

1.1 Meaning and scope of metal fabrication

Metal fabrication is the controlled process of converting engineering materials, particularly metals, into useful components, assemblies and structures. It begins with interpreting drawings and specifications and proceeds through material selection, measurement, marking out, cutting, forming, machining where necessary, joining, inspection, and finishing. Fabrication therefore covers much more than welding. Welding is one of the principal joining processes used within the wider fabrication sequence.

In construction and civil engineering, fabricated metal products include roof trusses, steel frames, handrails, staircases, gates, reinforcement cages, tanks, pipelines, bridge components, formwork systems, brackets, base plates and connection assemblies. The quality of these products depends on dimensional accuracy, correct material choice, suitable joint design, controlled heat input, competent workmanship and effective inspection.

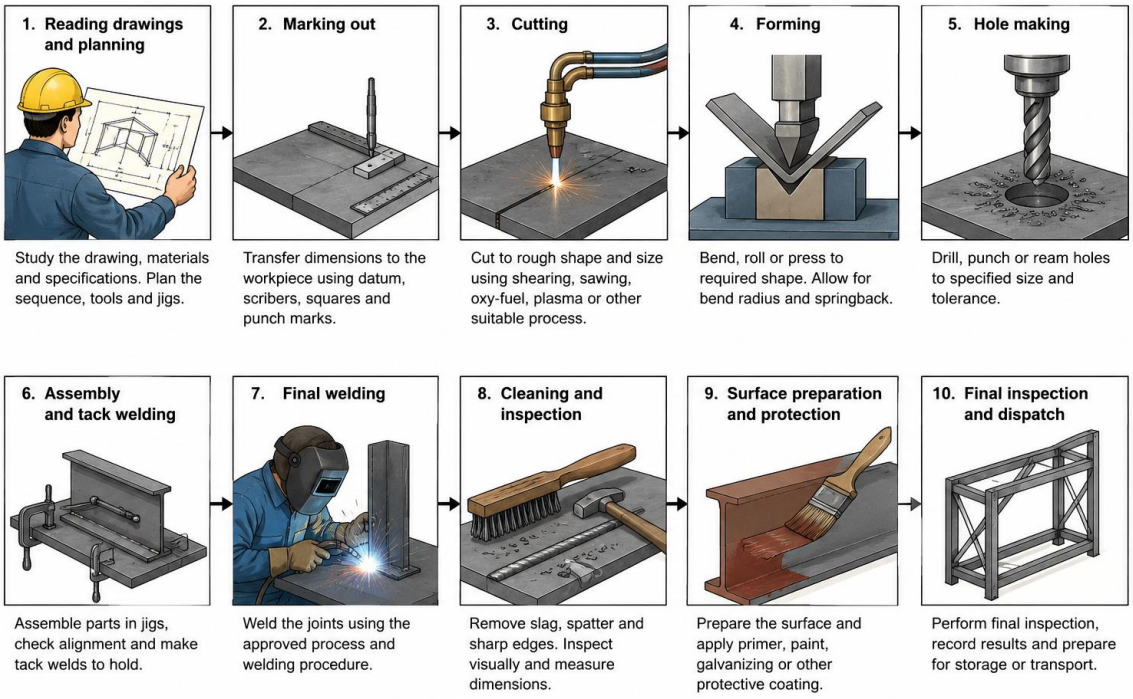


Figure 1.1: Typical sequence of operations in metal fabrication.

1.2 Meaning of welding

Welding is a permanent joining process in which two or more parts are united by the application of heat, pressure, or a combination of both, with or without filler metal. In fusion welding, the joint region is heated until the parent metal melts and a weld pool is formed. On cooling, the molten metal solidifies to create a metallurgical bond. In pressure welding processes, bonding may occur without extensive melting.

A welded joint is not merely a mechanical attachment. The heat cycle changes the microstructure and properties of the metal close to the joint. For this reason, welding must be understood as a thermal, metallurgical and structural process.

1.3 Advantages and limitations of welded construction

Advantages	Limitations / concerns
Produces strong, continuous and relatively light joints	Heat may cause residual stresses, distortion and metallurgical changes
Can join complex shapes and large structures	Quality can be sensitive to operator skill and procedure control
Provides leak-tight joints for tanks and pipes	Some defects are hidden and require non-destructive testing
Reduces the need for holes, bolts and overlapping plates	Repairs and disassembly are more difficult than bolted connections
Can be automated for high productivity	Requires attention to fumes, radiation, fire and electrical hazards

1.4 Learning outcomes

Explain the principles and terminology of welding and metal fabrication.

Identify common metals, welding processes, equipment and consumables.

Interpret basic welding symbols, joints and workshop drawings.

Select appropriate cutting, forming, welding and finishing processes.

Describe weld defects, inspection methods and quality-control procedures.

Estimate basic fabrication quantities, welding time and material requirements.

Apply safe workshop practices during welding, cutting and fabrication.

2. Engineering Materials for Welding and Fabrication

2.1 Classification of engineering metals

Engineering metals are broadly divided into ferrous and non-ferrous groups. Ferrous metals contain iron as the principal element and include carbon steels, alloy steels, cast irons and stainless steels. Non-ferrous metals include aluminium, copper, nickel, magnesium, titanium and their alloys. Each group behaves differently during heating, melting and cooling; therefore weldability cannot be judged from strength alone.

2.2 Carbon steels

Low-carbon, or mild, steel is the most widely fabricated structural metal because it combines adequate strength, ductility, availability, low cost, and good weldability. As carbon content increases, hardness and strength generally increase but weldability tends to decrease because the heat-affected zone becomes more susceptible to hardening and cracking.

Material	Typical characteristics	Fabrication implication
Low-carbon steel	Ductile, readily formed, generally good weldability	Common for structural frames, plates, pipes and general fabrication
Medium-carbon steel	Higher strength and hardness	May require preheating and controlled cooling
High-carbon steel	High hardness and wear resistance	Greater cracking risk; welding needs special procedures
Stainless steel	Corrosion resistant; chromium-rich	Requires suitable filler, cleanliness and heat control
Aluminium alloys	Low density; high thermal conductivity	Oxide removal and shielding are critical
Cast iron	High carbon; brittle	Repair welding commonly requires preheat and slow cooling

2.3 Weldability

Weldability is the capacity of a material to be welded under the imposed fabrication conditions into a structure that performs satisfactorily in service. It depends on chemical composition, thickness, restraint, welding process, heat input, joint geometry and service environment. A material may be physically weldable but still require special controls to avoid brittle heat-affected zones, hydrogen cracking, hot cracking or excessive distortion.

2.3.1 Carbon equivalent

For many structural steels, carbon equivalent is used as a practical indicator of hardenability and welding difficulty. A commonly used expression is $CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15$, where the element contents are expressed as mass percentages. As carbon equivalent increases, preheating and stricter hydrogen control become more important. Exact limits should be taken from the applicable welding code and steel specification.

2.4 Heat-affected zone

The heat-affected zone (HAZ) is the portion of the parent metal that does not melt but whose microstructure and mechanical properties are altered by the welding thermal cycle. The HAZ may contain regions of grain growth, transformation, tempering or softening. Its behaviour is particularly important in high-strength steels and heat-treated alloys.

3. Fundamentals of Welding

3.1 Formation of a welded joint

A sound fusion weld requires adequate energy to melt the joint faces, protection of the molten pool from harmful atmospheric gases, suitable filler metal where required, and controlled solidification. The energy source may be an electric arc, gas flame, electrical resistance, laser, electron beam or other source. During arc welding, electrical energy is converted into intense heat at the arc, where temperatures are sufficient to melt steel rapidly.

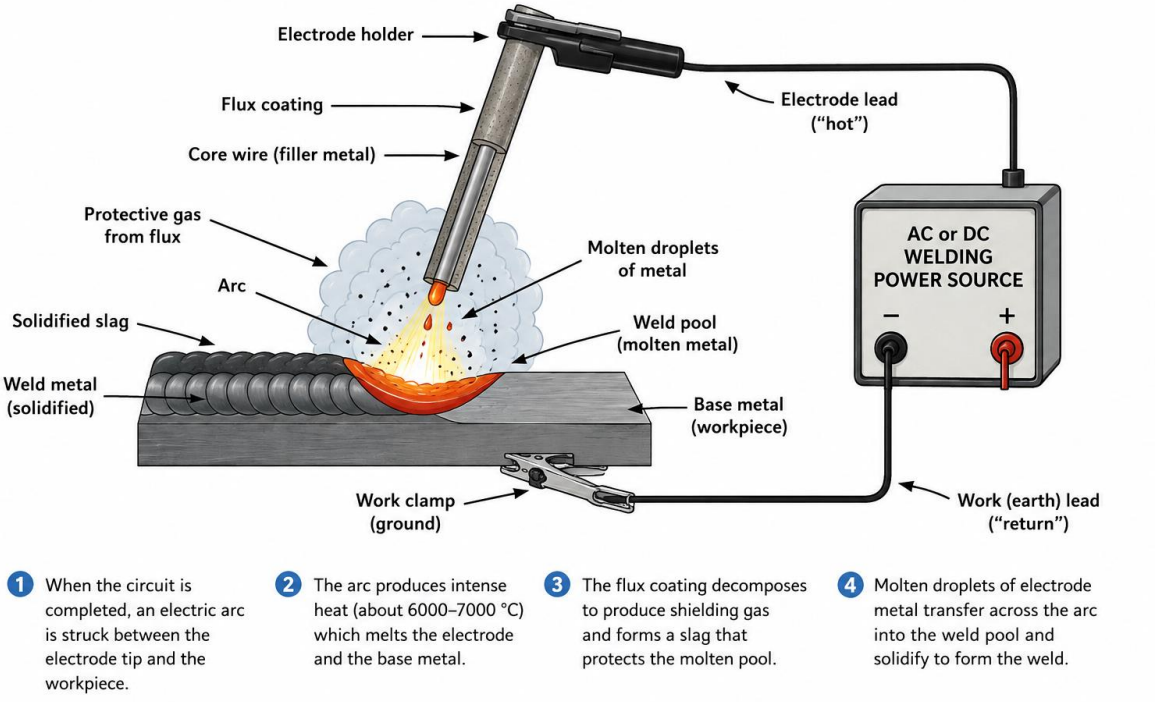


Figure 3.1: Simplified principle of electric arc welding.

3.2 Important welding terminology

Term	Meaning
Parent/base metal	The material being joined.
Filler metal	Metal added to form part of the completed weld.
Weld metal	Metal that has melted and resolidified in the weld.
Fusion face	Surface of the parent metal intended to be fused.
Root	The deepest part of the joint where members approach most closely.
Toe	Junction between the weld face and parent metal.
Weld face	Exposed surface of a weld on the side from which welding was carried out.
Penetration	Depth to which fusion extends into the parent metal.
Reinforcement	Weld metal extending above the surface profile of the joint.
Heat-affected zone	Parent metal altered by heat but not melted.

3.3 Heat input

Heat input is a useful measure of the thermal energy delivered per unit length of weld. For arc welding it is commonly related to voltage, current and travel speed. A simplified relationship is $H = \eta VI/v$, where H is effective heat input per unit length, η is process efficiency, V is arc voltage, I is welding current and v is travel speed, with consistent units. Excessive heat input may enlarge the HAZ and increase distortion, while insufficient heat input can cause lack of fusion or inadequate penetration.

Worked example 3.1: Relative heat input

A weld is deposited at 24 V and 180 A with a travel speed of 5 mm/s. Ignoring process efficiency for a simple comparison, the electrical energy per unit length is $VI/v = (24 \times 180)/5 = 864$ J/mm. If the travel speed is doubled while voltage and current remain unchanged, the energy per unit length is approximately halved. This demonstrates why slow travel generally produces a wider thermal cycle.

4. Welded Joints, Edge Preparation and Welding Positions

4.1 Common joint types

Joint design determines how forces are transferred through a welded assembly and strongly influences access, preparation cost, weld volume and inspection. The common basic configurations are butt, lap, T, corner and edge joints. Structural fabrication makes extensive use of butt and T-joints, with groove welds and fillet welds respectively.

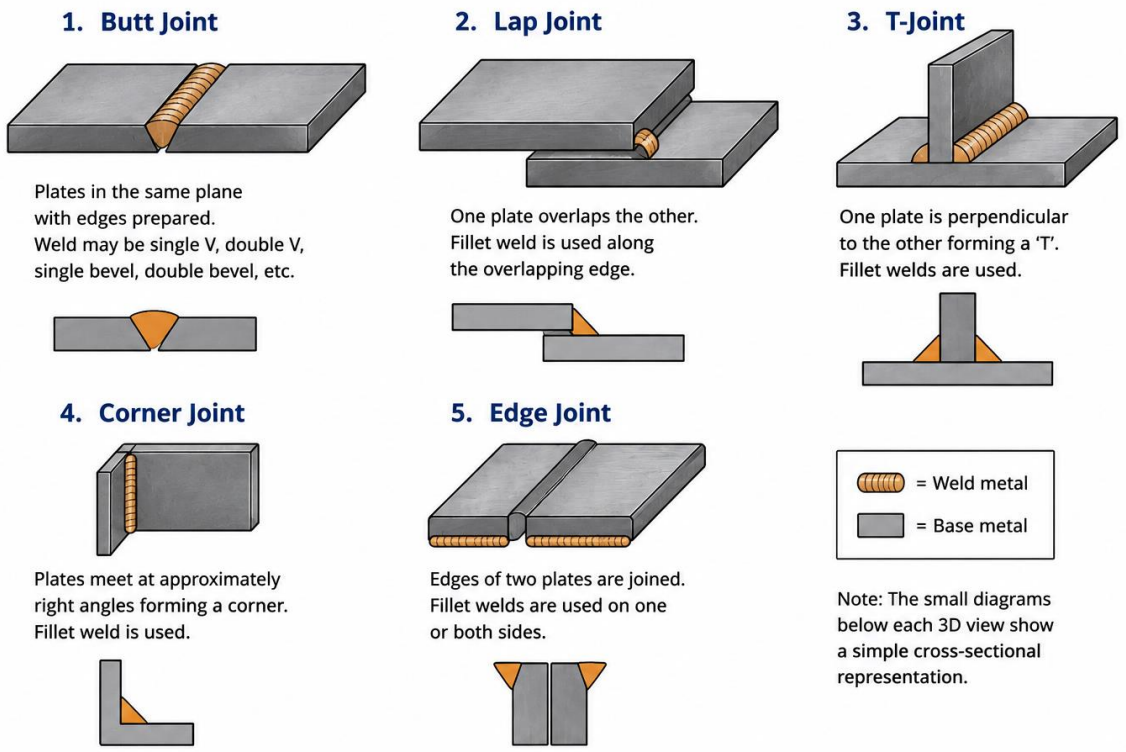


Figure 4.1: Simplified examples of common welded joint configurations.

4.2 Butt-joint preparations

Thin plates may be joined using square edges when adequate penetration can be obtained. Thicker plates generally require edge preparation so that the welding arc can reach the root and fusion can be achieved through the thickness. Typical preparations include single-V, double-V, single-bevel, double-bevel, U and J preparations. A double-sided preparation reduces weld metal volume and angular distortion for thick plates but requires access to both sides.

4.3 Fillet welds

A fillet weld has an approximately triangular cross-section and is commonly used in lap, T and corner joints. Important dimensions include leg length, throat thickness and weld length. In an ideal equal-leg fillet weld, the theoretical throat is approximately 0.707 times the leg size. Structural design must use the effective throat and applicable design code rather than simply the visible leg dimension.

4.4 Welding positions

Position	Description	Practical implication
Flat / downhand	Weld deposited from the upper side of a nearly horizontal joint	Easiest position; high deposition rates possible
Horizontal	Weld axis approximately horizontal on a vertical surface	Molten pool tends to sag downward
Vertical	Welding progresses vertically upward or downward	Requires careful manipulation and parameter control
Overhead	Weld deposited from underside of joint	Most demanding; pool control and PPE are critical

4.5 Fit-up and tack welding

Fit-up is the positioning and alignment of parts before final welding. Correct root gap, alignment, bevel angle and joint cleanliness are essential. Tack welds hold components in position and should be made using compatible consumables and adequate quality because defective tack welds can become part of the final joint or initiate cracking.

5. Shielded Metal Arc Welding (SMAW / Manual Metal Arc Welding)

5.1 Principle

Shielded Metal Arc Welding uses a flux-coated consumable electrode. An arc is established between the electrode tip and the workpiece. The electrode core wire melts and supplies filler metal, while the flux coating decomposes to produce shielding gases and a protective slag. The slag covers the hot weld metal and must normally be removed between passes and after completion.

5.2 Equipment

AC or DC welding power source.

Electrode holder and welding cable.

Work/earth return clamp and cable.

Flux-coated electrodes.

Chipping hammer and wire brush.

Welding helmet, gloves, protective clothing and screens.

5.3 Electrode selection

Electrodes are selected according to parent metal strength, welding position, power-source characteristics, penetration requirements, hydrogen control and service conditions. Low-

hydrogen electrodes are important for restrained joints and higher-strength steels because they reduce diffusible hydrogen in the weld. Such electrodes must be stored and handled according to the manufacturer's requirements to prevent moisture pickup.

5.4 Welding variables

Variable	Effect when increased, other factors constant
Current	Generally increases melting rate and penetration; excessive current causes spatter and undercut
Arc length	Higher voltage and wider arc; excessive length reduces shielding effectiveness
Travel speed	Higher speed reduces heat input and bead width; excessive speed may cause undercut/lack of fusion
Electrode angle	Controls arc force, pool support and bead placement
Electrode diameter	Larger electrodes permit higher current and deposition but require adequate joint access

5.5 Basic operating procedure

Study the drawing, welding procedure and safety requirements.

Prepare and clean the joint; remove paint, oil, rust and moisture near the weld.

Fit and tack the components to the required dimensions.

Select the correct electrode type, diameter, polarity and current.

Strike the arc and establish a stable arc length.

Maintain the required work angle, travel angle and travel speed.
Clean slag completely between passes and inspect each layer.
Complete the weld, remove slag and spatter, then conduct visual inspection.

6. Gas Welding and Oxy-Fuel Cutting

6.1 Oxy-fuel welding

Oxy-fuel welding produces heat by combustion of a fuel gas with oxygen. Oxy-acetylene welding is the best-known form because acetylene can produce a high-temperature flame suitable for welding many steels. The process is highly controllable and useful for thin sheet, repair work, brazing and heating, although it is slower than many arc processes for structural fabrication.

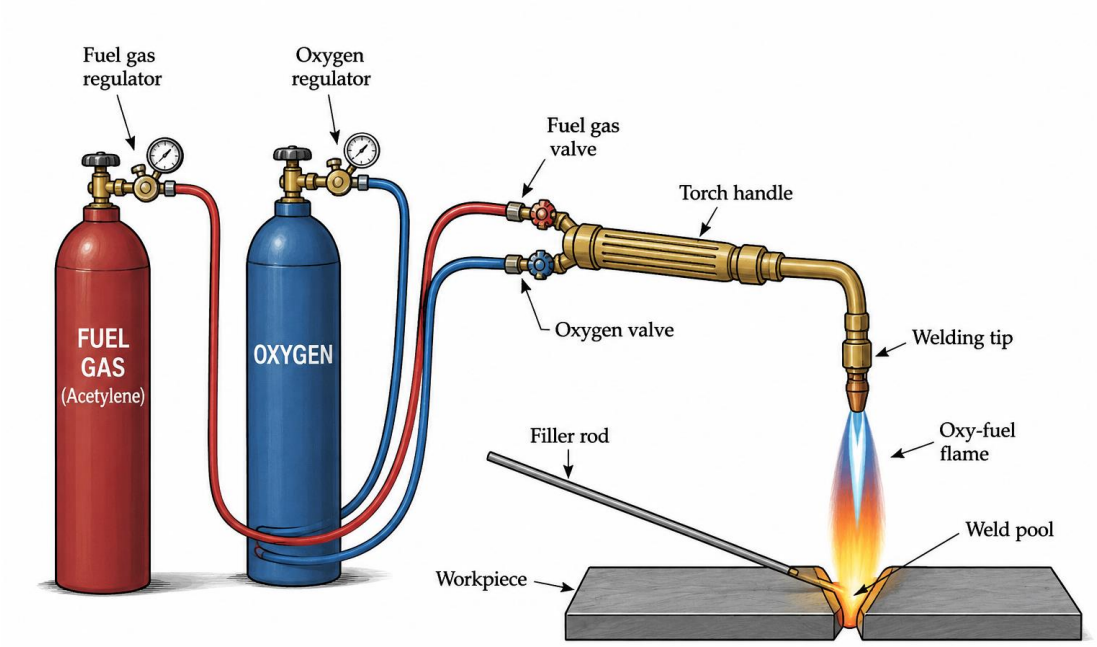


Figure 6.1: Simplified oxy-fuel welding arrangement.

6.2 Flame types

Flame	Gas balance	Typical appearance/use
Neutral	Approximately balanced oxygen and acetylene	General welding of many steels
Carburising/reducing	Excess acetylene	Reducing atmosphere; selected alloy applications
Oxidising	Excess oxygen	Short, sharp inner cone; useful for selected copper alloys and cutting-related heating

6.3 Oxy-fuel cutting

Oxy-fuel cutting is not simply melting the steel. The surface is first heated to ignition temperature, after which a high-purity oxygen jet reacts exothermically with iron and blows the resulting oxides from the kerf. The method is effective for carbon and low-alloy steels but is generally unsuitable for aluminium and stainless steel without specialised techniques because their oxides do not support the same cutting mechanism.

6.4 Cylinder safety

Gas cylinders must be secured upright, protected from heat and mechanical damage, and handled with correct regulators. Oxygen equipment must never be contaminated with oil or grease.

Flashback arrestors and non-return valves should be fitted where required. Leaks should be checked using an approved leak-detection method, never with a flame.

7. MIG/MAG, TIG and Flux-Cored Arc Welding

7.1 Gas Metal Arc Welding (GMAW: MIG/MAG)

Gas Metal Arc Welding feeds a continuous wire electrode through a welding gun. Shielding gas protects the arc and molten pool. The term MIG is commonly associated with inert shielding gases, while MAG uses active gases or mixtures. The continuous wire feed gives high productivity and makes the process suitable for mechanisation and production fabrication.

7.2 Gas Tungsten Arc Welding (GTAW/TIG)

TIG welding uses a non-consumable tungsten electrode to create the arc. Filler metal, when required, is added separately. An inert shielding gas, usually argon or an argon-based mixture, protects the electrode and weld pool. TIG offers excellent control and high-quality welds, especially on thin materials, stainless steel and aluminium, but deposition rates are relatively low.

7.3 Flux-Cored Arc Welding (FCAW)

FCAW uses a tubular wire containing fluxing ingredients. Some wires are self-shielded, while others use an external shielding gas. The process can achieve high deposition rates and is widely used in structural fabrication. Self-shielded variants can be advantageous outdoors where wind makes gas shielding difficult, subject to the approved welding procedure.

Process	Main electrode	Shielding	Typical strength
SMAW	Flux-coated stick electrode	Flux-derived gas and slag	Versatile, portable, suitable for site work
GMAW	Continuous solid wire	External gas	High productivity and easy automation
GTAW/TIG	Non-consumable tungsten	Inert gas	Excellent precision and weld quality
FCAW	Continuous flux-cored wire	Flux and/or external gas	High deposition and structural productivity

8. Resistance Welding, Brazing, Soldering and Other Joining Methods

8.1 Resistance welding

Resistance welding generates heat at the interface of contacting parts by passing a high electric current through the joint while applying pressure. Spot welding is widely used for sheet-metal

assemblies. The principal variables are welding current, electrode force and current duration. Because no filler metal is normally required, the process can be rapid and readily automated.

8.2 Brazing

Brazing joins materials using a filler metal that melts above approximately 450 °C but below the melting temperature of the parent materials. The parent metals do not melt. Capillary action draws molten filler into a close-fitting joint. Brazing is useful for joining dissimilar metals and delicate assemblies where lower thermal distortion is desirable.

8.3 Soldering

Soldering is similar in principle to brazing but uses lower-melting filler metals. It is common in electrical, electronic and light sheet-metal work. Soldered joints generally do not carry the high structural loads expected of welded joints.

8.4 Mechanical fastening

Bolts, screws, rivets and other mechanical fasteners remain important alternatives or complements to welding. Bolted joints are particularly useful where disassembly, field erection, tolerance accommodation or avoidance of welding heat is important.

9. Metal Fabrication Processes

9.1 Reading drawings and planning

Fabrication should begin from an approved drawing, bill of materials and specification. The fabricator must identify dimensions, tolerances, hole locations, weld symbols, material grades, surface finishes and assembly sequence. Planning should minimise unnecessary handling, reduce waste and ensure that welding access is available after components are assembled.

9.2 Measurement and marking out

Marking out transfers dimensions from the drawing to the workpiece. Common tools include steel rules, tapes, squares, scribers, dividers, punches, straight edges and templates. Accuracy depends on using a consistent datum and avoiding cumulative dimensioning errors. Where precision is critical, check measurements independently before cutting.

9.3 Cutting processes

Method	Best suited to	Key considerations
Hand hacksaw / powered saw	Bars, sections, pipes	Good dimensional control; slower for heavy plate
Shearing	Sheet and thin plate	Fast straight cuts; edge deformation possible
Oxy-fuel cutting	Carbon steel plate	Portable; creates HAZ and requires edge cleanup
Plasma cutting	Conductive metals	Fast, versatile and suitable for stainless/aluminium
Laser cutting	Precision sheet/plate work	High accuracy; high equipment cost
Grinding / abrasive cutting	Sections and finishing cuts	Flexible but produces sparks, dust and heat

9.4 Forming operations

Forming changes the shape of metal without intentionally removing material. Common operations include bending, rolling, folding, pressing and drawing. During bending, the outer fibres are stretched while inner fibres are compressed. Springback occurs when elastic recovery causes the bend to open slightly after the load is removed; tooling and bend allowance must therefore be considered.

9.5 Drilling and hole production

Holes may be drilled, punched, reamed or thermally cut depending on thickness, tolerance and specification. Structural bolt holes must comply with the relevant design and fabrication standard. Poorly located or oversized holes can create erection problems and reduce connection performance.

9.6 Assembly and jigs

Jigs, fixtures, strongbacks, clamps and temporary supports are used to hold components in the required geometry during assembly and welding. Good fixturing improves repeatability but excessive restraint can increase residual stress and cracking risk. The assembly sequence should balance dimensional control with freedom for thermal movement.

10. Welding Metallurgy and Thermal Effects

10.1 Heating and cooling cycle

Welding subjects a small region of metal to rapid heating followed by cooling through a steep temperature gradient. The weld metal solidifies from the fusion boundary toward the centreline, while adjacent parent metal experiences different peak temperatures depending on its distance from the fusion line. These thermal cycles determine grain size, phase transformation, hardness and toughness.

10.2 Residual stresses

When a weld cools, the hot weld metal and nearby HAZ attempt to contract. Surrounding cooler material restrains this contraction, producing self-equilibrating residual stresses. High tensile residual stress can contribute to brittle fracture, fatigue cracking and stress-corrosion problems under unfavourable conditions.

10.3 Distortion

Distortion is dimensional change caused by non-uniform expansion and contraction. Common forms include longitudinal shrinkage, transverse shrinkage, angular distortion, bowing and buckling. Thin plates and unsymmetrical welds are especially vulnerable.

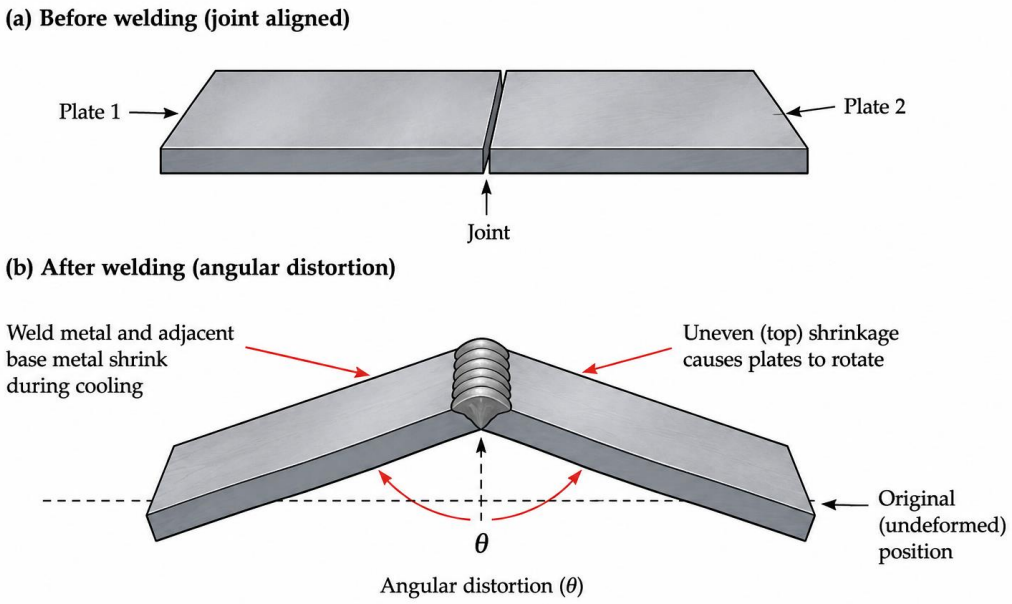


Figure 10.1: Simplified illustration of angular distortion caused by uneven weld shrinkage.

10.4 Control of distortion

Use the minimum weld size consistent with design requirements.

Adopt balanced or symmetrical welding where possible.

Use suitable welding sequences such as back-step or skip welding.

Preset components where predicted shrinkage is known.

Use clamps and fixtures carefully without creating excessive restraint.

Reduce unnecessary heat input and avoid overwelding.

Use double-sided joint preparations for thick plates where appropriate.

10.5 Preheating and post-weld heat treatment

Preheating raises the temperature of the parent metal before welding. It slows cooling, reduces temperature gradients and helps hydrogen diffuse away from susceptible regions. Post-weld heat treatment may be used to reduce residual stress, temper hardened microstructures or restore properties, depending on material and code requirements. These operations must follow an approved procedure because incorrect heating can damage the material.

11. Welding Symbols and Fabrication Drawings

11.1 Purpose of welding symbols

Welding symbols provide a compact method of communicating joint and weld requirements on engineering drawings. A complete symbol may include a reference line, arrow, basic weld symbol, dimensions, contour or finish information, supplementary symbols and a tail containing process or specification information.

11.2 Information commonly communicated

Type of weld, such as fillet or groove weld.

Weld size or groove depth.

Effective throat where specified.

Length and pitch of intermittent welds.

Side of joint to be welded.

All-around or field-weld requirement.

Contour and finishing method.

Applicable welding process or procedure reference.

Students should be aware that symbol conventions differ slightly among standards. In professional work, follow the drawing standard stated in the project documents consistently.

11.3 Fabrication tolerances

A tolerance defines the permissible variation from a specified dimension or geometry. Fabricated steelwork requires control of straightness, squareness, hole position, member length, plate flatness and assembly dimensions. Tolerances should not be invented by the workshop; they are taken from drawings, specifications and applicable fabrication standards.

12. Weld Defects, Discontinuities and Their Prevention

12.1 Defect versus discontinuity

A discontinuity is an interruption in the normal structure or geometry of a weld. A discontinuity becomes a defect when it exceeds the acceptance criteria of the applicable code or specification. This distinction is important because not every visible imperfection automatically makes a weld unacceptable.

Discontinuity	Typical causes	Prevention / control
Porosity	Moisture, contamination, inadequate shielding, long arc	Clean joint, dry consumables, correct gas flow and arc length
Slag inclusion	Poor cleaning, narrow groove, incorrect manipulation	Remove slag between passes; provide access and correct technique
Lack of fusion	Low heat input, fast travel, poor electrode angle	Increase effective heat; improve technique and joint preparation
Incomplete penetration	Insufficient root opening/heat; poor preparation	Correct root geometry and welding parameters
Undercut	Excess current, long arc, excessive travel speed	Adjust current, arc length, travel and electrode angle
Cracks	Hydrogen, hard HAZ, high restraint, solidification problems	Control consumables, preheat where required, reduce restraint, correct filler
Overlap	Low travel speed or poor angle	Correct manipulation and parameter balance
Burn-through	Excess heat on thin material or excessive root gap	Reduce current/heat input; improve fit-up and backing

12.2 Cracking

Cracks are among the most serious welding imperfections because they can propagate under service loading. Hot cracking occurs at high temperature during solidification or shortly afterward, while hydrogen-assisted cold cracking may occur after the weld has cooled. Cold cracking is promoted by susceptible microstructure, tensile stress and diffusible hydrogen; controlling all three factors is therefore central to prevention.

13. Inspection, Testing and Quality Assurance

13.1 Quality assurance in fabrication

Quality assurance is the planned system used to ensure that fabricated work meets specified requirements. It includes qualified procedures, competent personnel, calibrated equipment, material traceability, consumable control, inspection plans, dimensional checks and documented acceptance.

13.2 Visual inspection

Visual testing is the first and most widely used inspection method. It should be performed before, during and after welding. Before welding, the inspector checks joint preparation, cleanliness, alignment and consumables. During welding, parameters, interpass cleaning and sequence are monitored. After welding, the weld profile, size, surface cracks, undercut, overlap, arc strikes and dimensional distortion are assessed.

13.3 Non-destructive testing

Method	Principle	Typical application / limitation
Dye penetrant testing	Liquid penetrant enters surface-breaking discontinuities	Non-porous materials; detects only surface-breaking flaws
Magnetic particle testing	Magnetic flux leakage reveals surface/near-surface flaws	Ferromagnetic materials only
Ultrasonic testing	High-frequency sound reflects from internal interfaces	Internal defects; requires skilled interpretation
Radiographic testing	X-rays or gamma rays create an image through thickness	Good volumetric record; radiation safety is essential
Visual testing	Direct or aided observation	Fast and economical; limited to accessible surfaces

13.4 Destructive tests

Destructive tests are used for procedure qualification, welder qualification, material assessment and investigation. Examples include tensile tests, bend tests, impact tests, hardness tests, macro-etch examinations and fracture tests. Test selection depends on the performance characteristic that must be demonstrated.

13.5 Welding Procedure Specification

A Welding Procedure Specification (WPS) is a controlled document that defines how a weld is to be made. It typically states the welding process, joint design, material group, filler metal, position, preheat, electrical variables, travel requirements and other essential variables. In code-controlled work, the WPS is supported by qualification evidence such as a Procedure Qualification Record.

14. Welding and Fabrication Safety

14.1 General safety philosophy

Welding and fabrication combine electrical energy, intense radiation, hot metal, compressed gases, fumes, noise, rotating machinery and fire hazards. Safety therefore requires hazard identification before work starts, suitable engineering controls, correct PPE, competent supervision and disciplined housekeeping.

14.2 Major hazards and controls

Hazard	Examples	Key controls
Arc radiation	UV and infrared exposure to eyes/skin	Correct welding helmet shade, screens, covered skin
Fumes and gases	Metal fumes, shielding gases, coatings	Source extraction, ventilation, remove hazardous coatings, respiratory controls where required
Electric shock	Damaged leads, wet conditions, live parts	Inspect equipment, keep dry, isolate before maintenance, correct grounding/return
Fire/explosion	Sparks, hot slag, flammable materials	Hot-work permit, remove combustibles, fire watch, extinguishers
Compressed gases	Cylinder rupture, leaks, flashback	Secure cylinders, correct regulators, leak checks, flashback protection
Grinding/cutting	Flying particles, wheel failure, noise	Guards, correct discs, face/eye protection, hearing protection
Manual handling	Heavy plates and sections	Mechanical lifting, team lifting, correct slinging

14.3 Personal protective equipment

Typical PPE includes a welding helmet with appropriate filter lens, safety glasses, leather gloves, flame-resistant clothing, safety footwear and hearing protection. Respiratory protection may be necessary when ventilation alone cannot adequately control exposure. PPE is the final layer of defence and should not replace ventilation, guarding or safe work methods.

14.4 Confined spaces

Welding in tanks, vessels, pits or other confined spaces requires special controls because fumes, oxygen deficiency, gas accumulation and restricted escape can become fatal. Atmospheric testing, forced ventilation, isolation of connected services, standby arrangements and a formal confined-space procedure may be required.

15. Fabrication of Structural Steel Components

15.1 Structural fabrication sequence

Structural steel fabrication commonly proceeds through material receipt and identification, cutting to length, drilling or punching, plate preparation, assembly, welding, dimensional inspection, surface preparation, coating and dispatch. Traceability is important where material grades and test certificates must be linked to the finished component.

15.2 Columns, beams and trusses

Beams and columns may require end plates, stiffeners, base plates, cleats and splice components. Trusses demand accurate jigging because small dimensional errors at individual joints can accumulate and cause erection difficulties. Welding sequence should be planned to limit bowing, twisting and angular distortion.

15.3 Base plates and anchor interfaces

Base plates transfer column forces into concrete foundations through bearing, anchors and sometimes grout. Fabrication must control plate flatness, hole location, column position and weld geometry. Incorrect hole spacing can prevent erection even when the welds themselves are sound.

15.4 Corrosion protection

After fabrication and inspection, steel surfaces may be protected by painting, hot-dip galvanising, metallising or other coating systems. Surface preparation is critical because mill scale, rust, oil and welding residues reduce coating performance. Weld spatter and sharp edges should be treated as required by the coating specification.

16. Pipes, Tanks and Sheet-Metal Fabrication

16.1 Pipe fabrication

Pipe fabrication involves cutting, bevel preparation, alignment, tack welding and circumferential welding. Internal misalignment, root gap and ovality must be controlled because they affect root

penetration and flow conditions. Pipe welding often uses multiple passes: root, hot pass, fill passes and cap.

16.2 Tank fabrication

Tanks require leak-tight joints and careful control of plate distortion. Shell courses may be assembled from rolled plates, while floor and roof plates are fitted and welded in planned sequences. Inspection may include visual testing, vacuum-box testing of floor seams, penetrant testing, magnetic testing or volumetric methods depending on design and code requirements.

16.3 Sheet-metal work

Sheet-metal fabrication includes shearing, notching, folding, bending, rolling, seaming, riveting, spot welding and light-gauge arc welding. Because thin sheet has low thermal mass, excessive welding heat easily causes burn-through and buckling. Intermittent welding, short welds and suitable fixtures are often used to control distortion.

17. Workshop Calculations and Estimating

17.1 Material take-off

A fabrication estimate begins by extracting component sizes and quantities from drawings. Lengths, plate areas and section masses are calculated and allowances are added for cutting loss, offcuts and procurement sizes. Standard section mass per metre is normally obtained from the manufacturer or steel tables.

Worked example 17.1: Mass of a steel plate

For a rectangular steel plate 1.20 m long, 0.60 m wide and 10 mm thick, volume = $1.20 \times 0.60 \times 0.010 = 0.0072\ m^3$. Using a steel density of approximately 7850 kg/m³, mass ≈ $0.0072 \times 7850 = 56.5$ kg. This value excludes cutting waste.

17.2 Approximate fillet-weld metal volume

For an equal-leg fillet weld of leg size s, the ideal triangular cross-sectional area is approximately $s^2/2$. The theoretical weld volume is this area multiplied by weld length. Actual deposited metal will differ because of weld profile, penetration and reinforcement, so production estimating normally applies suitable factors based on procedure and experience.

Worked example 17.2: Idealised fillet weld

For a 6 mm equal-leg fillet weld 1,000 mm long, ideal triangular area = $6^2/2 = 18\ mm^2$. Idealised volume = $18 \times 1{,}000 = 18{,}000\ mm^3 = 18\ cm^3$. If steel density is taken as 7.85 g/cm³, the

idealised metal mass is about 141 g before allowing for profile, deposition efficiency and wastage.

17.3 Welding time

Arc-on time depends on weld length and travel speed. If 2.4 m of weld is deposited at 4 mm/s, arc-on time = 2400/4 = 600 s = 10 min. Actual production time is longer because it includes electrode or wire changes, slag removal, repositioning, inspection and other delays. The ratio of arc-on time to total working time is often called operating factor or duty utilisation in estimating contexts.

17.4 Cost components

Parent metal and purchased sections.

Filler metal, electrodes, wire, flux and shielding gas.

Labour for preparation, fit-up, welding, cleaning and inspection.

Power, fuel gas and equipment operating cost.

Consumables such as grinding discs and cutting tips.

Surface preparation and protective coatings.

Non-destructive testing and quality documentation.

Overheads, handling, transport and fabrication waste.

18. Practical Workshop Exercises

18.1 Exercise A: Marking, cutting and squaring a mild-steel plate

Students should interpret a simple drawing, establish a datum, mark the required dimensions, centre-punch reference points, cut the plate using an approved method, remove burrs and verify final dimensions. Assessment should consider dimensional accuracy, edge quality, tool handling, housekeeping and PPE.

18.2 Exercise B: SMAW bead-on-plate

The purpose is to develop arc striking, arc-length control, travel speed and electrode-angle control. Students should deposit parallel beads on a cleaned mild-steel plate, remove slag and compare bead width, reinforcement, ripple uniformity and evidence of undercut or porosity.

18.3 Exercise C: Fillet-welded T-joint

Prepare two plates, tack them at approximately 90 degrees, verify squareness and deposit a fillet weld of specified size. After cooling and cleaning, inspect both weld profile and assembly distortion. Where facilities permit, the specimen can be sectioned and macro-etched to observe penetration and fusion.

18.4 Exercise D: Oxy-fuel cutting

Students should inspect the equipment, set regulators under supervision, light and adjust the flame correctly, preheat the steel edge and initiate the oxygen cutting jet. The cut is evaluated for straightness, kerf width, drag lines, dross and heat damage.

18.5 Suggested workshop record

Item	Student record
Drawing/specification	Sketch and dimensions of specimen
Material	Grade/type and thickness
Process	Welding/cutting process used
Consumable	Electrode/wire/gas details
Parameters	Current, voltage where available, polarity, travel observations
Inspection	Observed weld profile and discontinuities
Safety	PPE and controls applied
Reflection	Causes of errors and proposed improvement

19. Process Selection for Engineering Applications

19.1 Selection criteria

No single welding process is best for every job. Selection should consider material type, thickness, joint design, required quality, welding position, site conditions, production volume, accessibility, available power, operator competence, equipment cost and inspection requirements.

Application	Often suitable processes	Reasoning
General site structural steel	SMAW, self-shielded FCAW	Portable and tolerant of site conditions
Workshop structural production	GMAW/MAG, FCAW, submerged arc where applicable	High productivity and mechanisation potential
Thin stainless sheet	TIG or controlled GMAW	Heat control and clean weld appearance

Aluminium fabrication	TIG or MIG with suitable equipment	Effective inert-gas shielding and suitable filler
Precision pipe root welding	TIG or approved specialised process	High control of root quality
Sheet-metal mass production	Resistance spot welding	Rapid cycle and automation

19.2 Engineering decision-making

Process selection should be treated as an engineering decision rather than a matter of operator preference. A process that deposits metal rapidly may still be unsuitable if it cannot achieve the required toughness, access, positional capability or environmental control. The final choice should satisfy the design specification and the qualified welding procedure.

20. Maintenance, Repair and Good Workshop Practice

20.1 Repair welding

Repair welding begins with diagnosis. The damaged region must be inspected to determine the material, defect extent, loading condition and likely cause of failure. Cracks should be completely removed before rewelding, and the repair procedure should address preheat, filler metal, sequence and inspection. Simply welding over a crack can conceal rather than correct the defect.

20.2 Equipment care

Inspect welding leads, electrode holders, torches, earth clamps, regulators, hoses, and power sources regularly. Loose electrical connections create resistance heating and unstable arcs. Gas leaks and damaged hoses require immediate correction. Grinding machines must use compatible, undamaged discs with guards in place.

20.3 Housekeeping

A well-organised fabrication shop reduces accidents and improves quality. Store offcuts safely, route cables and hoses to reduce trip hazards, clearly identify hot metal, separate flammable materials from hot work, and return tools to designated locations.

21. Summary, Review Questions and Examination Practice

21.1 Chapter summary

Welding is a controlled metallurgical joining process embedded within the broader discipline of metal fabrication. Successful fabrication requires correct interpretation of drawings, suitable materials, accurate preparation, appropriate process selection, controlled welding parameters, dimensional management, inspection and safe working practice. The most important engineering principle is that weld quality is created during planning and production; inspection alone cannot manufacture quality after the weld has been completed.

21.2 Review questions

Differentiate welding from metal fabrication and give four construction-related examples of fabricated products.
Explain why low-carbon steel is generally easier to weld than high-carbon steel.
Describe the heat-affected zone and explain why it is important.
Compare SMAW, GMAW/MAG, GTAW/TIG and FCAW in terms of electrode, shielding and typical use.
Explain the functions of electrode flux in SMAW.
Sketch and describe butt, lap and T-joints.
Explain the relationship between current, travel speed and heat input.
Discuss five common weld discontinuities, their causes and preventive measures.
Explain how welding distortion develops and describe six methods of control.
Compare visual, penetrant, magnetic particle, ultrasonic and radiographic testing.
Outline a safe procedure for oxy-fuel equipment setup and shutdown.
Explain the purpose of a Welding Procedure Specification.
Describe a logical fabrication sequence for a welded structural steel frame.
Calculate the approximate mass of a 2.0 m × 0.5 m × 8 mm steel plate using 7850 kg/m³.
Discuss factors that should govern the selection of a welding process for a construction project.

21.3 Suggested examination problems

Problem 1: A fabrication shop must produce repeated 8 mm mild-steel fillet-welded frames under controlled indoor conditions. Compare SMAW and GMAW/MAG and recommend a process based on productivity, equipment, shielding and operator considerations.
Problem 2: A welded plate assembly shows severe angular distortion after welding. Explain the thermal mechanism responsible and propose a revised joint preparation, welding sequence and restraint strategy.
Problem 3: During visual inspection, a weld shows undercut, excessive spatter and irregular bead width. Identify likely parameter or technique problems and propose corrective actions.
Problem 4: A restrained medium-carbon steel joint is to be welded in a cool workshop. Discuss why preheat, low-hydrogen consumables and controlled heat input may be specified.

Glossary of Key Terms

Term	Definition
Arc	Electrical discharge between electrode and workpiece that produces welding heat.
Backing	Material placed at the root to support molten weld metal or assist penetration.
Bead	A single deposit of weld metal resulting from one welding pass.
Crater	Depression at the termination of a weld bead.
Deposition rate	Mass of weld metal deposited per unit time.
Duty cycle	Permissible operating time of welding equipment at a stated output over a specified cycle.
Filler metal	Metal added during welding, brazing or soldering.

Flux	Material used to protect, clean or modify the weld region during joining.
HAZ	Heat-affected zone: parent metal altered by welding heat without melting.
Interpass temperature	Temperature of the weld region immediately before the next pass.
Kerf	Width of material removed by a cutting process.
Pass	Single progression of welding along a joint.
Polarity	Electrical relationship between electrode and workpiece in DC welding.
Preheat	Heating of the base material before welding.
Root gap	Separation between joint members at the root.
Slag	Non-metallic solid product formed from flux and impurities.
Tack weld	Short weld used to hold parts in alignment before final welding.
Undercut	Groove melted into parent metal adjacent to the weld toe and left unfilled.
Weld pool	Local volume of molten metal before solidification.
WPS	Welding Procedure Specification.

Suggested Further Reading

American Welding Society (AWS). Welding Handbook. Current edition.

American Welding Society (AWS). Structural Welding Code—Steel (AWS D1.1/D1.1M). Current edition.

International Organisation for Standardisation (ISO). Relevant standards for welding quality, personnel qualification and weld imperfections.

Jeffus, L. Welding: Principles and Applications. Current edition.

Kou, S. Welding Metallurgy. Current edition.

Kalpakjian, S., & Schmid, S. Manufacturing Engineering and Technology. Current edition.

Relevant national occupational safety regulations, project specifications and manufacturers' welding equipment manuals.